Design a video streaming platform like Youtube. — watch, upload } on demand!!

① Requirements / Scope

- Functional Requirements
  - Watch
    - (a) watch video on demand
    - (b) to start watching video from time last watched
    - (c) search for video (keyword based search
  - upload
    - (a) Upload video
  - View & add
    - comments
    - likes.

- Non-functional
  - Scalable
  - Available / consistent } highly available
  - Reliable / Trustworthy
  - Latency should be low.

② Estimating the infrastructure

- Total no. of users = 2 bn (50% - active users ↳ 1 bn)

→ Youtube
- viewers
- content-creators

} No. of views → 400 X No. of videos uploads.

On an avg - each user watches = 4 videos

∴ Total no. of reads = 1 X 4 bn videos = 4 B/day

$$= \frac{4B}{86400} /s$$

$= 50$ videos $/s$.
$\times 10^3$

No. of views $=$ 400 $\times$ No. of writes

$\Rightarrow$ No of writes $= \frac{\text{No. of views}}{400} = \frac{50 \times 10^3}{400} = 125$ uploads/s

<u>Storage Needed</u> (Based on videos uploaded)

↳ Every minute ~ 400 hours worth data

~ 1 min (100 MB of storage) $\longrightarrow$ $400 \times 100 \times 60$ to min

$\longrightarrow 24 \times 10^5$ MB

$\rightarrow$ 2400 GB

$\rightarrow$ 2.4 TB /min

$\rightarrow$ 2.4 TB/min

$$= \frac{2.4}{60 \times 10} = 40 \text{ GB/s}$$

High level Design

Two flows
- Upload
- Viewing

Upload

(Can be asynchronous)

C → uploading

↳ should be able to check status of upload.

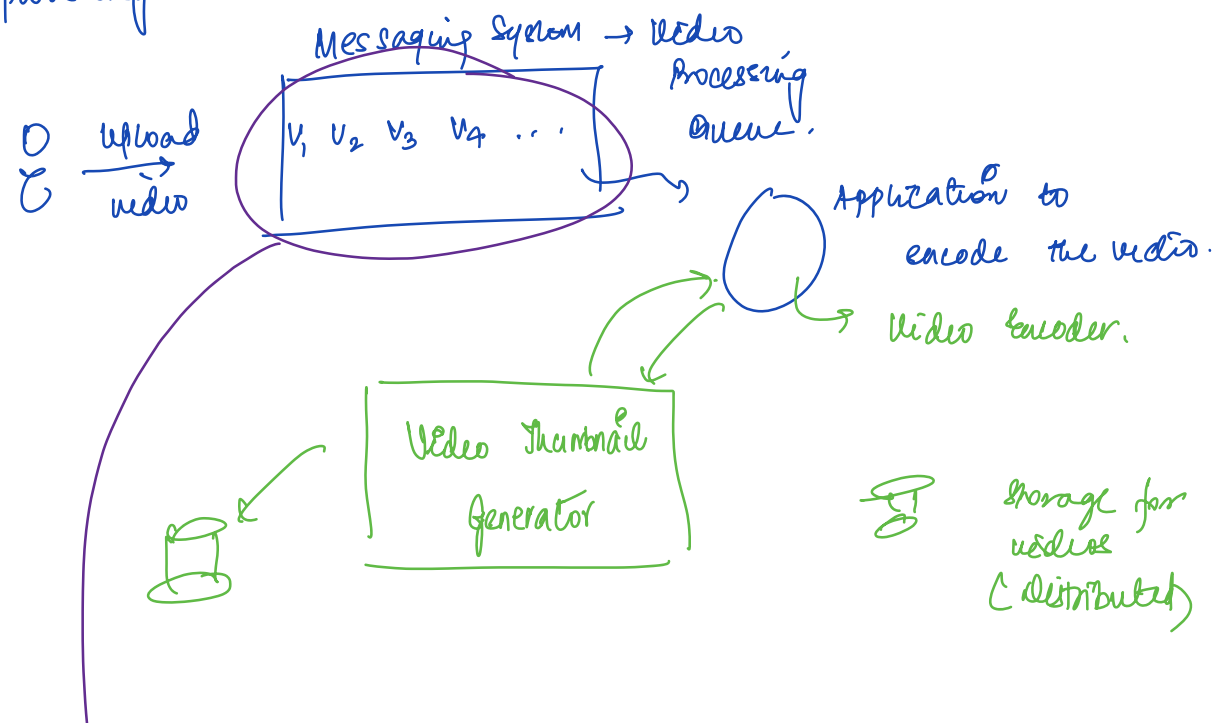
Viewing

(a) Able to search for video.

(b) Able to watch video

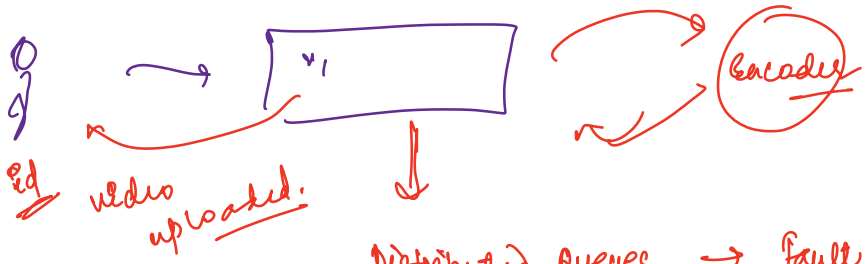
(c) Maintain offset of video based on user & device.

Uploading flow →



Video Processing Queue

[Queue] → FIFO



Distributed Queues → Fault Tolerance.

Video Encoder

Acts like consumer to the processing queue

distributed to scale well.

messaging system

Video Thumbnail generator → distributed

Each video will have multiple thumbnails

[More data]

Mongo DB.
[No SQL database]

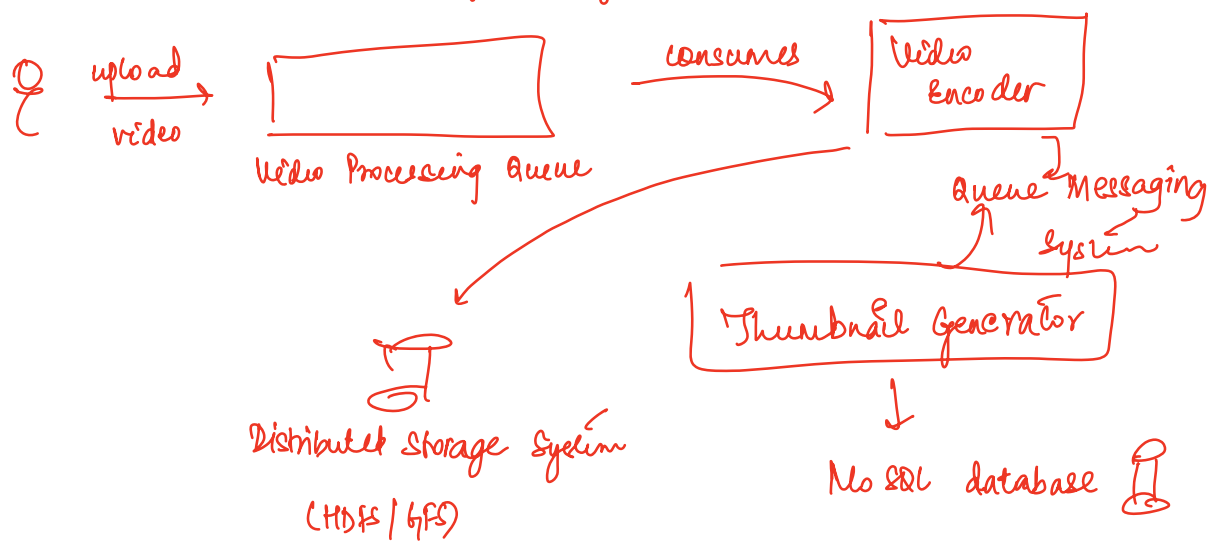
storage for thumbnails

Video Storage

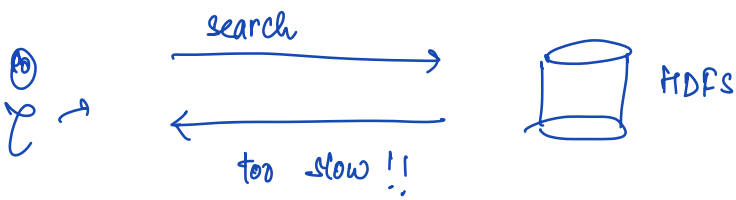
- Rate of storage of data = 40 GB/s (We've to keep adding data)

  ↳ HDFS / GFS (Google File System)

  # Ensure replication (Fault Tolerance)



View the video

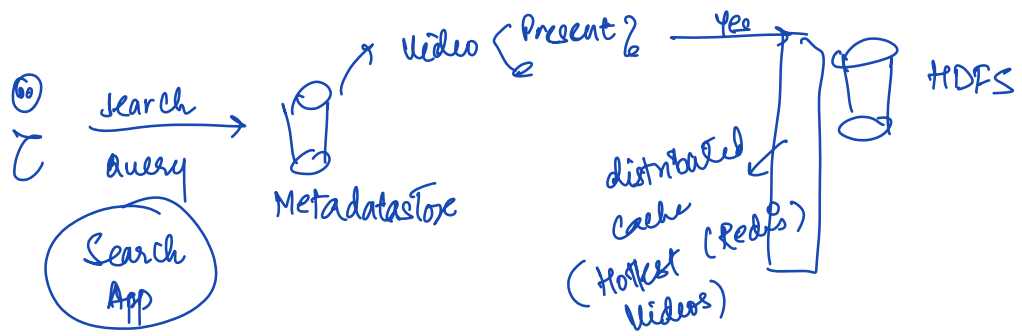


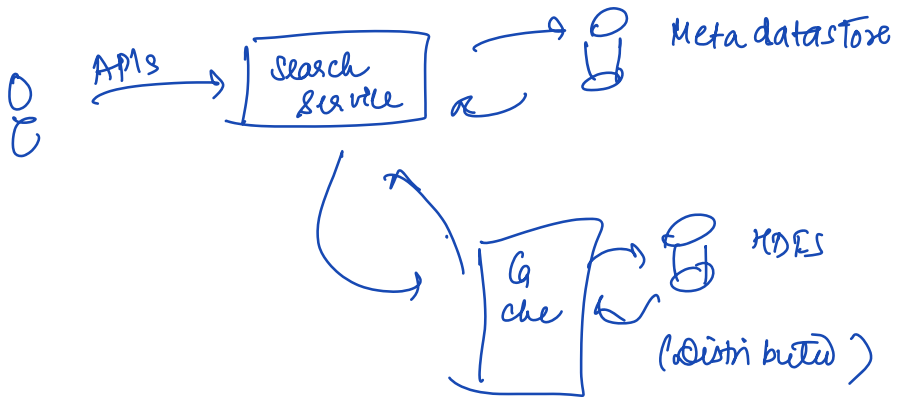
[Video Metadatastore]

↳ RDBMS Metastore

Name
Tags
Comments
Key words
Likes

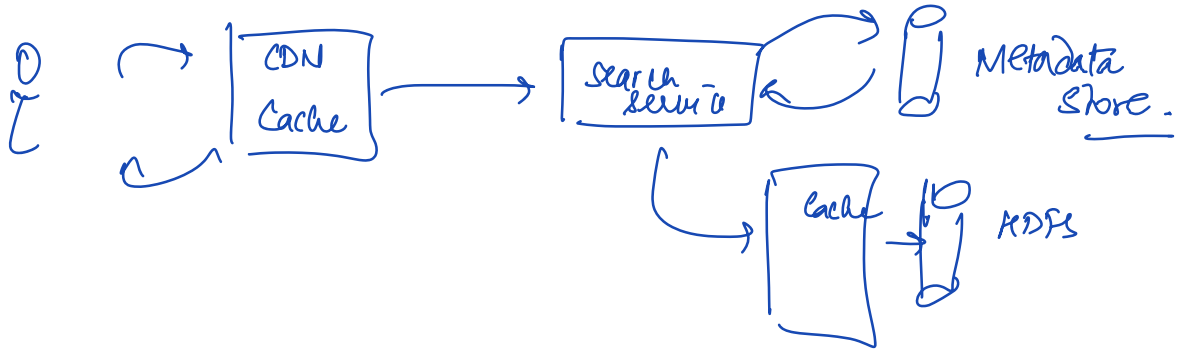
offset

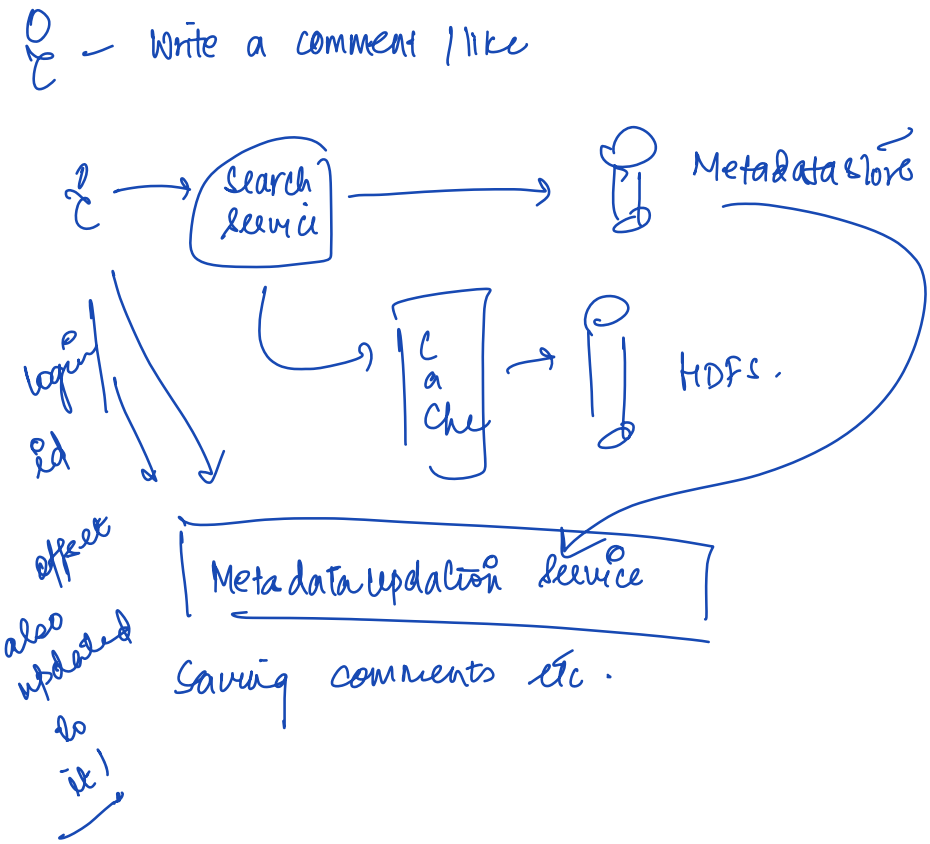




Improving Experience

for Caching → can use CDN (videos are static content)




Write a comment / like
Search service
Metadata store
Cache
HDFS.
login id
effect also updated to it!
Metadata updation service
Saving comments etc.